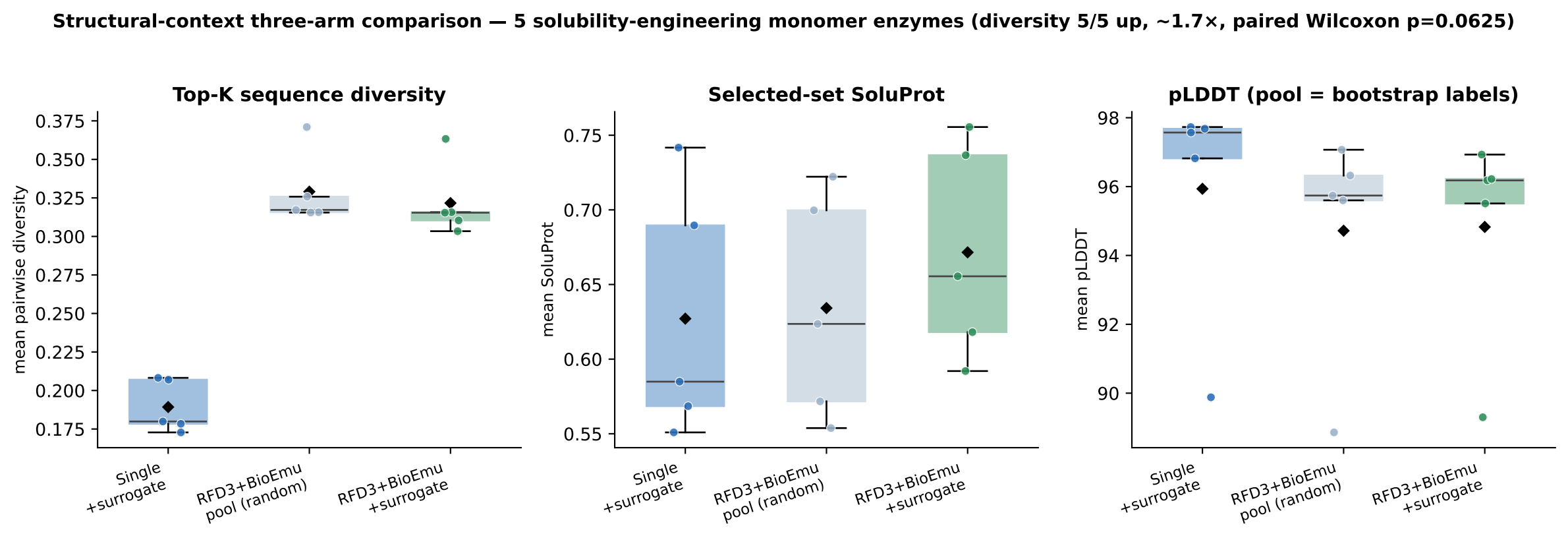
Structural-context three-arm comparison — 5 solubility-engineering monomer enzymes (diversity 5/5 up, ~1.7×, paired Wilcoxon p=0.0625)
Top-K sequence diversity
mean pairwise diversity
0.375
0.350
0.325
0.300
0.275
0.250
0.225
0.200
0.175
Single +surrogate
RFD3+BioEmu pool (random)
RFD3+BioEmu +surrogate
Selected-set SoluProt
mean SoluProt
0.75
0.70
0.65
0.60
0.55
Single +surrogate
RFD3+BioEmu pool (random)
RFD3+BioEmu +surrogate
pLDDT (pool = bootstrap labels)
mean pLDDT
98
96
94
92
90
Single +surrogate
RFD3+BioEmu pool (random)
RFD3+BioEmu +surrogate